

Geometry-agnostic data-driven thermal modeling of additive manufacturing processes using graph neural networks

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ARTICLE INFO

Keywords:

Additive manufacturing
Machine learning
Graph neural network

ABSTRACT

Additive manufacturing (AM) is commonly used to produce builds with complex geometries. Despite recent advancements in data-driven modeling of AM processes, the generalizability of such models across a wide range of geometries has remained a challenge. Here, a graph-based representation using neural networks is proposed to capture spatiotemporal dependencies of thermal responses in AM processes. Our results tested on the Directed Energy Deposition process, indicate that our deep learning architecture accurately predicts long thermal histories for unseen geometries in the training process, offering a viable alternative to expensive computational mechanics or experimental solutions.

1. Introduction

Additive manufacturing (AM) processes are increasingly utilized in a variety of industries such as aerospace, medical and automotive industries due to their superior capabilities to produce high-value-added parts with complex geometrical features and limited material consumption [1,2]. Despite AM's immense potential, its widespread adoption is hampered by challenges such as quality variability and process inefficiencies. For example, Glerum et al. [3] show that various material properties result from the same set of process parameters. Therefore, many state-of-the-art research attempts aim to establish a direct mapping between AM process parameters and resulting part properties [4]. The thermal profile during AM processing, resulting from process parameters, is a key signature connecting process parameters to defects (e. g., lack of fusion, distortion, hot cracking), melt pool characteristics, and consequently build properties. However, mapping the thermal profile and the process parameters based on experimental data is not just expensive, due to the high cost of material and equipment, but more importantly, incomplete due to the infeasibility of measuring experimental thermal profiles at every location of a part. Thus, thermal profile prediction in AM processes is essential in meeting the ever-increasing requirements for AM part quality and reliability.

Physics-based modeling has been the dominant approach to capture the thermal behavior of AM processes. High-fidelity computational models such as the finite element method (FEM) and computational fluid dynamics (CFD) offer high accuracy; however, they require

massive computational resources and time, making them infeasible for most time-sensitive applications. Therefore, many research attempts investigated alternative approaches to thermal modeling in AM processes. Empirical solutions based on experimental data are proposed to model melt pool temperature for real-time prediction and control systems [5,6]. Yet, they offer low accuracy as they over-rely on limited experimental data and fail to capture interconnected dependencies in process parameters. Alternatively, many analytical and semi-analytical solutions are proposed to offer a trade-off between the computation cost and the accuracy of high-fidelity models. While early versions of such analytical solutions were only applicable to single-track scenarios, recent publications extend their capability to multi-track and multi-layer cases [7–9]. Despite recent progress in the field, analytical solutions deviate from realistic responses even in moderately complex geometries as they include numerous simplifying assumptions.

Recently, data-driven modeling has shown the potential to contribute to the design, discovery, diagnosis, and optimization of advanced manufacturing processes [10,11]. In our previous work [12], it is demonstrated that data-driven methods for thermal profile prediction in AM offer a viable alternative to physics-based models. We developed a recurrent neural network (RNN) based architecture to predict the thermal histories of arbitrary points in AM builds. It is shown that the RNN architecture predicts the behavior of samples with similar classes of geometries to the ones in the training database with a scaled mean-squared-error of $3e^{-5}$; however, the error can significantly increase in cases for unseen classes of geometries. A surrogate model was

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<https://doi.org/10.1016/j.addma.2021.102449>

Received 5 March 2021; Received in revised form 17 October 2021; Accepted 26 October 2021

Available online 6 November 2021

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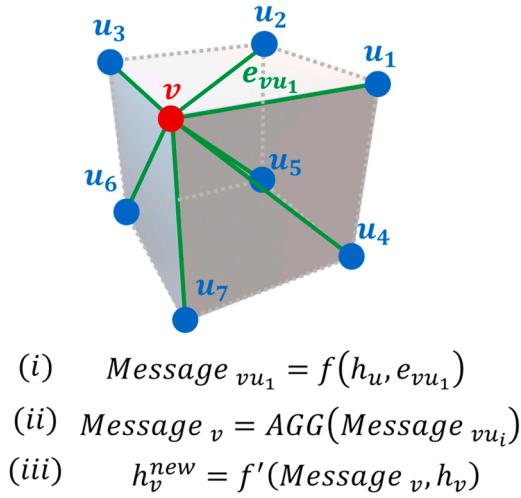


Fig. 1. Schematic of a target node and its neighboring nodes within an element. Message passing includes three fundamental steps: (i) message construction, (ii) message aggregations, and (iii) target update.

proposed by Roy et al. [13] to achieve a real-time thermal history prediction. A set of input features (e.g., distance from heat source and cooling surface) are selected from the geometry representations using GCode to reduce the computational demands. Their method can be generalized to different sizes of similar geometries, various process parameters, and different materials achieving a prediction accuracy of 95%. Two machine learning-based models using extreme gradient boosting (XGBoost) and long short-term memory (LSTM) are proposed by Zhang et al. [14] to predict the thermal histories using six input variables, including laser power, scan speed, layer index, time index, average height, and width. Models are trained and validated using real-time experimental measurements taken by IR thermal cameras under setups with varying process parameters. The authors reported a best runtime of 0.34 s for XGBoost, which is small enough for real-time prediction of data captured by IR cameras. A framework combining an RNN and a deep neural network (DNN) is developed by Ren et al. [15] to establish a mapping between the thermal history distributions and laser scanning patterns in the laser directed deposition process. The RNN-DNN model, trained with numerical simulation results analyzed by finite element methods as the ground truth, achieves accuracies higher than 95% but it can only predict thermal behaviors for single-layer rather than multilayer or complex geometries, which highly limits the application of the proposed algorithm. In the study by Haghighi et al. [16], physics-based and data-driven approaches are combined to characterize the filament bonding and the porosity distribution in an extrusion-based AM process for PLA (a thermoplastic polyester material). An analytical heat transfer model was applied for thermal profile characterization and an artificial neural network was adopted for filament deformation characterization. Their hybrid model achieved an average accuracy of 95% and 94% in predicting inter-layer and intra-layer bonding, respectively. In summary, however, the aforementioned data-driven approaches demonstrate their results only for simple geometries, such as thin walls and cubic structures.

As one can see, despite major achievements in data-driven modeling, state-of-the-art approaches fall short on the key issue of generalization across unseen geometries, which is crucial for AM modeling as AM is mostly used for producing one-of-a-kind complex geometries. The objective of this work is to introduce a novel physics-aware data-driven

thermal model for AM processes that can benefit from the high predictive power and computational efficiency of an artificial intelligence-empowered model while drastically improving its ability to generalize across challenging geometries. Our approach captures the intricacies of the physics through graph modeling, which provides a flexible representation of complex unstructured geometries. Additionally, the approach follows the local contributions of each node to other nodes within an element (in our case, a hexahedron connecting 8 nodes) and provides fundamentally similar calculation pathways to primary physics-based approaches, i.e., FEM. We demonstrate our approach on the Directed Energy Deposition (DED) process. DED is one of seven major AM families [17] where metal powders are blown into a melt pool generated from a laser or electron beam. Note that although the methodology is demonstrated for DED, it can be easily extended across many element-based predictive modeling approaches in applied physics and engineering. In what follows, first, the key elements of our methodology are introduced in Section 2, including our proposed architecture (Section 2.1) and database generation approach (Section 2.2). The results are discussed in Section 3 and we conclude this paper with our final remarks and future research directions in Section 4.

2. Graph-based modeling methodology

Graph neural networks (GNN) are deep learning architectures that capture dependencies in unstructured graphs via message passing between neighboring nodes. Due to their flexibility, GNNs have rapidly gained popularity in social sciences [18], chemistry [19], and image processing [20]. The two major categories of GNN include spectral-based and spatial-based methods. Spectral-based methods (e.g., GCN [21], AGCN [22], CHEBNET [23]) define a convolution operation based on the Laplacian eigen-basis, which makes this class most applicable to problems with a static graph representation [24]. In contrast, spatial-based GNNs define the convolution operation on the graph by aggregating the information from neighboring nodes and edges. Many researchers have successfully extracted spatial and temporal dependencies in traffic flow forecasting by considering traffic networks as graphs and integrating a GNN with a RNN (e.g., ST-GCN [25], TGC-LSTM [26], GGRU [27]). Spatial-based GNN formulations consist of three steps: (i) message creation and propagation at the starting nodes, (ii) message aggregation at the target nodes, and (iii) updating calculations based on the aggregated message. Here, two nodes are considered neighbors if they belong to the same element. A schematic of our definition of neighboring nodes and the three message-passing steps are demonstrated in Fig. 1.

In this work, we hypothesize that the spatiotemporal dependencies in AM processes which are traditionally modeled via physics-based simulations, such as FEM, can be alternatively captured using graph neural networks. This hypothesis stems from the resemblance between finite element matrix assembly operations (which combine the local contributions of element interconnectivity) and graph neural network message-passing formulations. To test this hypothesis, a database of simulation-based thermal responses is developed while building a variety of industrial-grade AM parts. We investigate two GNN architectures to reproduce AM responses. The schematics of the two architectures are depicted in Fig. 2 and their details are elaborated upon in the following subsections. Additionally, detailed block diagrams of the networks and a schematic of the training process along with hyperparameters is provided in Appendix A - Figs. 6 and 7.

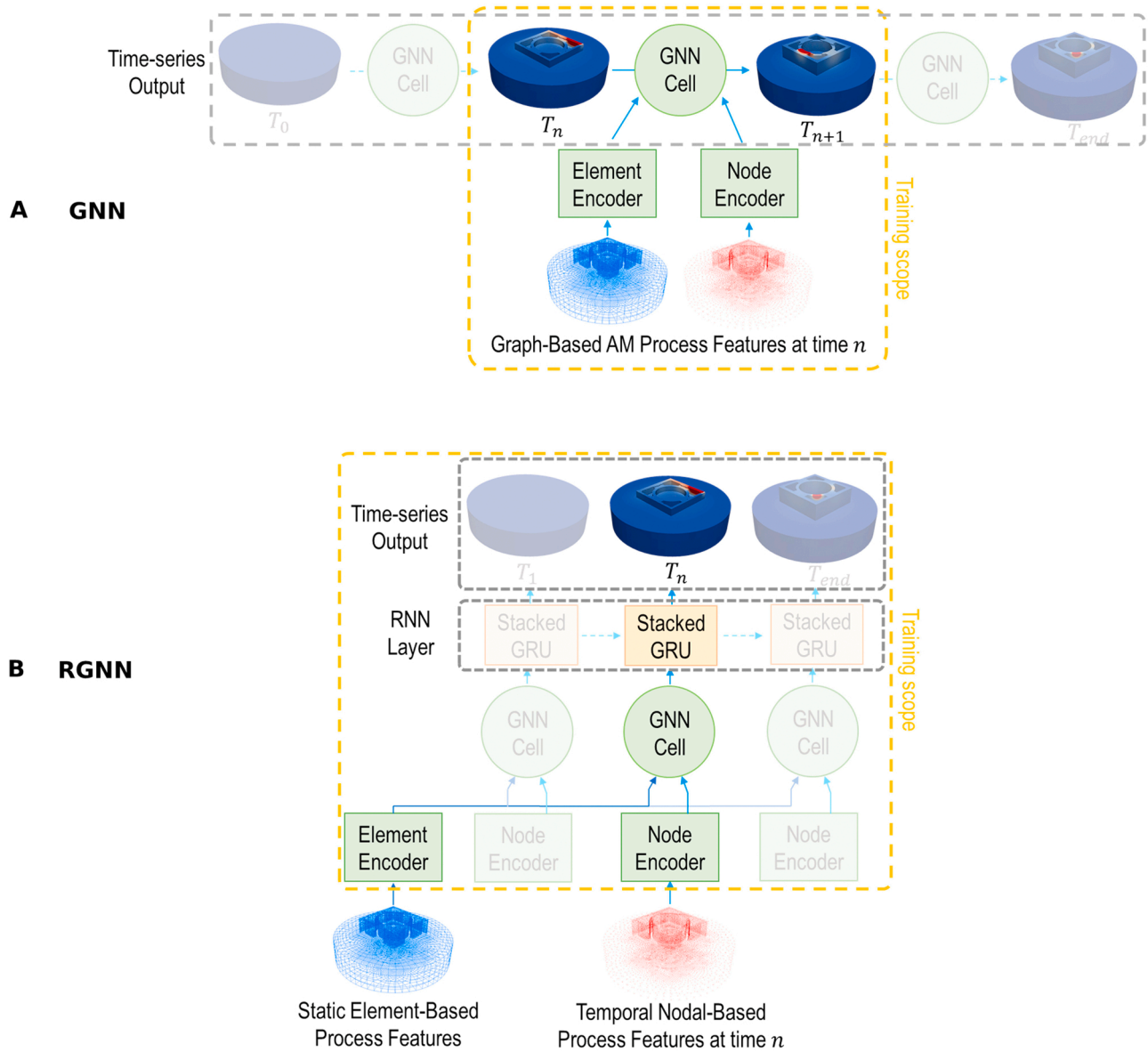


Fig. 2. Schematics of the two architectures for spatiotemporal prediction of AM thermal responses: (A) The GNN architecture predicts the single-time step update in each training instance given the node and element features at the time-step; (B) The RGNN architecture predicts and trains multi-time step interactions where at each time step the network receives a temporal nodal-based encoded representation, a non-temporal element-based representation, and the hidden state of the previous stacked GRU cell and outputs the thermal distribution over the geometry. Both architectures can be recursively evaluated to produce thermal outputs of arbitrary length.

Table 1

Process and material properties for the generated database. Material properties are obtained from [32].

Material Properties (SS316L)		Process and Environmental Properties	
Density	8000 Kg/m ³	Ambient Temperature	300 K
Heat Capacity	0.5 J/g K	Laser Power	1 KW
Latent Heat	272.5 J/g	Laser Diameter	2 mm
Conductivity Coefficient	21.4 W/m K	Report time step	0.1 s
Solidus/Liquidus	1648.15/1673.15 K	Scan speed	10 mm/s

2.1. Proposed network architectures

2.1.1. GNN architecture

As the first option, a GNN-based architecture is devised in which the network receives the thermal responses in the previous time step as well as process parameters at that time including nodal and element-based features (detailed in Section 2.2), and outputs the thermal responses in the current time step, as depicted in Fig. 2A. In essence, the network is responsible for calculating thermal fluxes at each time step to properly

update the thermal fields. In this network, each training or evaluation step predicts one forward time increment, however, by recursively providing the network output as the next time increment's input we can produce time-series responses of arbitrary length. For the GNN cell formulations in Fig. 2A, we found that DeeperGCN [28] empirically leads to better results compared to existing alternatives, though it is noteworthy that in our experience the difference in performance has been insignificant. In the DeeperGCN formulation, h_v^l (node features for the l th layer) will be passed to a layer-normalization layer, an activation layer with a rectified linear unit, and a drop-out layer for feature pre-processing. Subsequently, a spatial convolution is performed with the preprocessed features to update the features for the next layer. The spatial convolution operation is defined as [28]:

$$h_v^{(l+1)} = MLP(h_v^{(l)} + AGG(\{ReLU(h_u^{(l)} + e_{vu}^{(l)}) + \varepsilon : u \in \mathcal{N}(v)\})) + h_v^{(l)} \quad (1)$$

where $\mathcal{N}(v)$ is the set of neighboring nodes of v , $h_u^{(l)}$ represents the neighbor node features and $e_{vu}^{(l)}$ represents the features of edges connecting v and u . $ReLU$ is the rectified linear unit activation function and AGG is the aggregation function, which is the Softmax function in the current work. MLP is a multi-layer perception and ε is a small positive constant to assure numerical stability. As shown in Eq. (1), the encoded

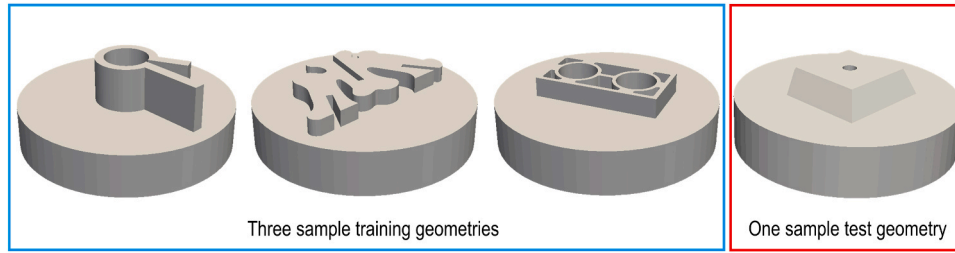


Fig. 3. Sample AM builds adapted from the ABC online repository [33] for industrial-grade geometries. Geometries are oriented and placed on substrate plates to construct the AM simulation database. The three geometries within the blue border are in the training dataset while the geometry within the red border is used as one of the test samples. All geometries are provided in the Appendix B - Fig. 8.

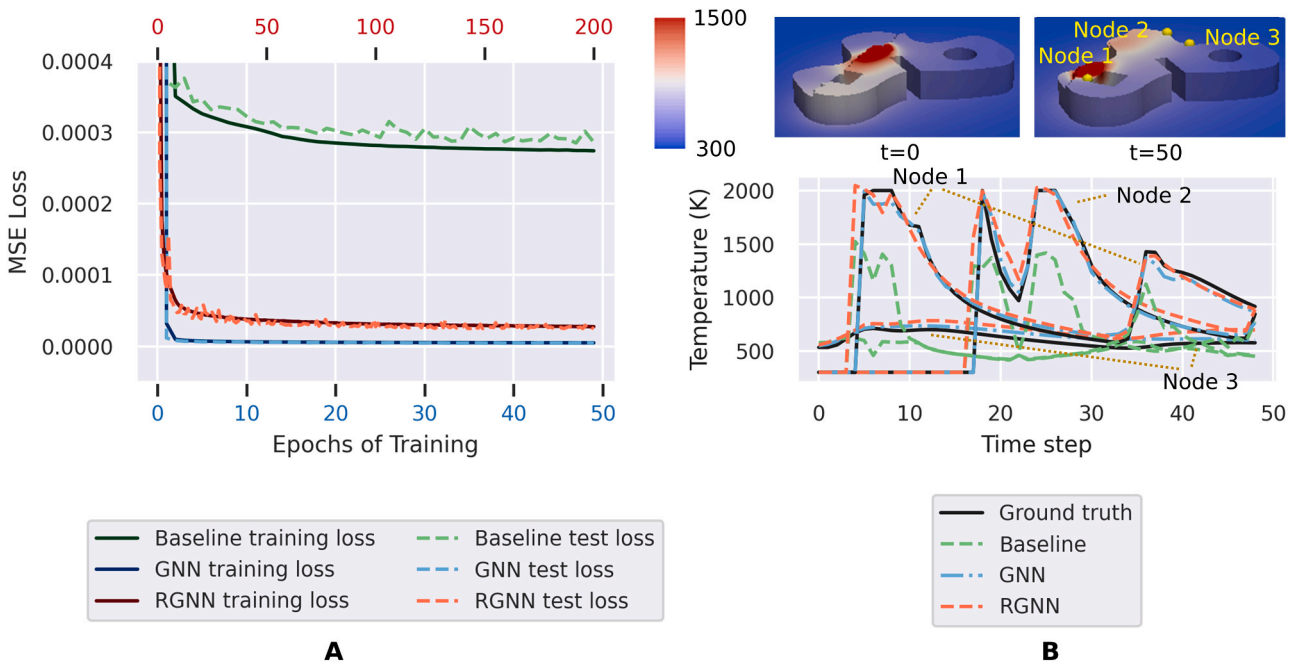


Fig. 4. Training and evaluation results for the baseline, GNN and RGNN formulations: (A) The evolution of the train and test losses over training epochs is normalized per node per time step; (B) An example simulation and the predicted thermal history at three points with the location of points depicted on the top right and the comparison of histories between baseline, GNN, RGNN and the ground truth on the lower right. Note that $t = 0$ refers to the starting time of the 50 time-step test sample and not the entire build.

Table 2

Summary of MSE results for baseline, GNN and RGNN models on training and test and the demonstration case in Fig. 4B. Additionally, the errors are reported as the percentage of the baseline performance in parentheses. Note that for training and test sets reported MSE correspond to each model's training span, i. e., 50 time steps for RGNN and one time step for baseline and GNN. The demonstration case results are provided for 50 time step for all three models by recursively unrolling the predictions.

	Training set MSE (% of baseline solution)	Test set MSE (% of baseline solution)	Demonstration case MSE in Fig. 4B (% of baseline solution)
Baseline	$2.74e^{-4}$ (100%)	$2.86e^{-4}$ (100%)	$4.49e^{-4}$ (100%)
GNN	$4.88e^{-6}$ (1.78%)	$4.36e^{-6}$ (1.52%)	$3.57e^{-5}$ (7.95%)
RGNN	$2.75e^{-5}$ (10.0%)	$2.47e^{-5}$ (8.63%)	$5.32e^{-5}$ (11.8%)

neighboring node features $h_u^{(l)}$ and edge features $e_{vu}^{(l)}$ are first added and fed to an activation function to construct the message from each neighboring node. The message is aggregated and then combined with nodal features in an *MLP* to calculate an update. Finally, the updates are added to the previous node to construct the output of the GNN layer. The last step provides the residual connection which not only facilitates the network training process, but also improves the interpretability of the models because it resembles thermal fluxes.

2.1.2. RGNN architecture

As an alternative architecture, to potentially better model long

temporal dependencies of the thermal system, we developed a Recurrent Graph Neural Network (RGNN) architecture as depicted in Fig. 2B. In this architecture, the network takes the time-series nodal features and edge features as the inputs and directly generates time-series thermal responses over an arbitrary number of time steps. At each time step, a GNN cell (like the cell used in Section 2.1.1) is used to capture local interactions between nodes and elements using their corresponding features. By concatenating the candidate updates generated by the GNN cells with shared parameters, a time-series candidate update (c_t) is assembled, which feeds into a stacked RNN layer. For the RNN formulation, the Gated Recurrent Unit (GRU) [29] is adopted in this work as follows:

$$r_t = \text{sig}(W_r \cdot [s_{t-1}, c_t] + b_r) \quad (2)$$

$$z_t = \text{sig}(W_z \cdot [s_{t-1}, c_t] + b_z) \quad (3)$$

$$\hat{s}_t = \tanh(W \cdot [r_t \times s_{t-1}, c_t] + b) \quad (4)$$

Table 3

Summary of RMSE results for GNN and RGNN models unrolled over 1000 time steps.

	Training set RMSE	Test set RMSE	Demonstration case RMSE in Fig. 5
GNN	$1.50e^{-2}$	$1.44e^{-2}$	$1.47e^{-2}$
RGNN	$9.58e^{-3}$	$9.32e^{-3}$	$8.53e^{-3}$

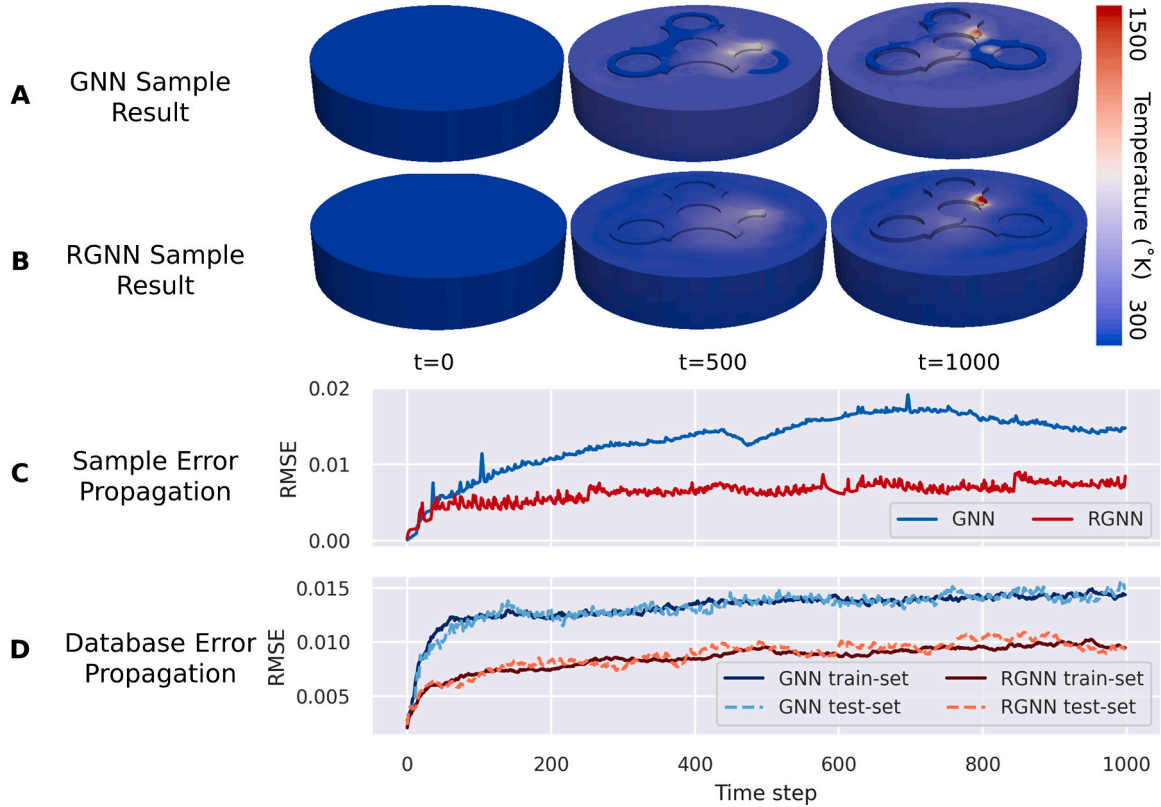


Fig. 5. Evaluation of the trained models' capability to produce long-term simulations. The evolution of the thermal field on a sample simulation is depicted for the GNN and RGNN models (A and B). The error propagation of the sample simulation and all database simulations for both models are shown (C and D).

$$T_t = s_t = (1 - z_t) \times s_{t-1} + z_t \times \hat{s}_t \quad (5)$$

where GRU cell takes a candidate update (c_t) and the hidden state s_{t-1} as the input, and outputs the temperature field at the current time, T_t . r_t and z_t represent the reset and update gates, correspondingly, where each uses a separate sets of weights (W), biases (b), and a sigmoid activation function. The output (T_t) is used as the hidden state for the next time step (s_t) and the hidden state is initialized with the initial temperature field T_0 . Further details on both networks can be found in [Appendix A](#) as well as our publicly available code.¹

2.2. AM database development

We developed a database based on high-fidelity finite element simulations, which has allowed us to have access to the thermal histories of all geometric points. An explicit in-house AM simulation package, GAMMA [30,31], is used to solve the transient heat transfer equations while we gradually activate elements as the toolpath passes over a predefined mesh. Heat conduction, convection, radiation, and external heat flux as the result of the laser beam, are modeled in our simulations, while stainless steel 316 L is used as the material. Key process and material properties are reported in [Table 1](#). In principle, one can add variations to the process and material properties and train the network to generalize over a range of such parameters (e.g., as in [12]). However, here, the parameters provided in [Table 1](#) are kept constant to focus on the capability of the model to predict challenging geometry and toolpath-related features.

To ensure that the proposed models are trained and tested on diverse geometries, we selected 55 different industrial-grade geometries from the ABC database, where 45 of them are used for training and 10 geometries are randomly separated for testing. Four samples of the geometries are shown in [Fig. 3](#), while the complete set of geometries is provided in [Appendix B - Fig. 8](#). The selected geometries vary in their size, the number of layers, shapes, and geometric features (e.g., wall thickness) to capture a wide range of AM builds. CAD geometries are scaled and placed on a substrate of 20 mm height and 100 mm diameter to make them suitable for sample DED manufacturing. The geometries are meshed using ABAQUS with 8-node hexahedron elements, where each element has an approximate edge size of 5 mm for the substrate and 1 mm for the part, resulting in about 10–100k elements in the meshes. To incentivize the network to generalize across different toolpaths, a Python script is developed to automatically generate layer-by-layer toolpaths for arbitrary geometries, where the contour patterns are randomly selected from 9 toolpath strategies varying in their motion directions, patterns (zigzag versus spiral), and starting positions. Due to computational restrictions, we limit this study to geometries that can be deposited on a substrate with a 100 mm diameter. Statistics on the geometric feature sizes in the database are provided in [Appendix B - Fig. 9](#).

The database is created using graph representations, where the graph typology is constructed based on the meshed geometry, i.e., every node of the mesh is defined as a node in the graph, and the edges are defined according to the connectivity matrix that indicates which nodes are within a common element. Each node of the graph is embedded with three features: (i) birth flag (indicating whether a node is active at that time step), (ii) layer height, and (iii) laser distance feature defined as the inverse of the distance between each node and the laser beam at each time step. In addition to matrix connectivity, we consider an element-based distance feature indicating the distances between any two nodes of an element. Examples of the input features are provided in [Appendix A - Figs. 6 and 7](#). The three nodal features and one element-based feature are selected to provide the barebone information

necessary about the boundary conditions and fluxes to the data-driven model. The laser distance determines the external flux input into the system, the layer height indicates the distance from the boundary condition (the bottom of the substrate), and finally the birth flag and edge distance determine the neighboring interactions between nodes. Only minimal feature engineering is applied (e.g., inverting the laser distance), as it is expected that the data-driven network automatically extracts hidden correlations between input features.

As element connectivity and feature are constant in time, we implement them as static inputs to the model across all time steps, whereas nodal features are provided as time-series inputs (see [Fig. 2B](#)). The two networks require different sampling methods. For the GNN architecture, we randomly sample pairs of inputs (process features and thermal responses at time n) and outputs (thermal responses at time $n + 1$) at different times of the simulations. In contrast, the RGNN model receives time-series inputs and outputs of 50 consequent time steps (equivalent to 25 cm of deposition), where the starting times are sampled from the length of the simulations. To provide a fair comparison between the performances of the two networks, they train over the same amount of data collected from the same simulations split into two sets of training and test data. All inputs and outputs of the models are normalized to values between 0–1 to assist the optimization process. While cross validation methods can often increase the accuracy of final model, we design this analysis based on two completely separated datasets of training and test sets because it allows for a better analysis of the generalization capabilities of the model. Additionally, note that the computational cost of cross validation during our training is prohibitively high.

There are several scientific and practical merits in our data-driven solution, even if it is trained on a synthetic database. First, once trained, the presented data-driven solution offers orders of magnitude faster evaluation compared to FEM simulations (see the last paragraph in [Section 3](#)). Therefore, the model has practical uses in many time-sensitive applications. Second, we believe investigating the capability of the model with synthesized data is an essential scientific step before adopting an experimental database because it allows us to isolate, study, and resolve important computational problems. Third, in the most realistic scenarios, it is conceivable that the ultimate solution for data-driven modeling involves both experimental and simulation data as they can complement each other's features and augment the size of the database. Therefore, the capability of models to work with simulation data is an important topic of research.

3. Results and discussion

The two architectures are implemented using the PyTorch deep learning library [34] and PyTorch Geometric package [35] in Python. The models are trained on their corresponding training sets, while the test sets are only deployed for evaluation without contributing to backpropagation. Each training epoch consists of a complete pass on training samples in which we update model parameters and simultaneously calculate a mean squared error (MSE) of the predictions versus database ideal responses. After each epoch, all test samples are evaluated and their MSE errors are stored. We stop each training process when no improvements in the MSE of the test samples are seen.

We also introduce a baseline solution, which is the same as the developed GNN architecture, only without the previous temperature as the input. The reason behind this choice is two-fold. First, training a GNN model without access to its past allows us to evaluate the time-dependent nature of the problem. If a data-driven model without temporal input can lead to a reasonable performance, the modeling approach can be greatly simplified. Second, a baseline allows for a more intuitive assessment of the proposed models compared to raw error

¹ https://github.com/AMPL-NU/Graph_AM_Modeling

metrics.

Our results, depicted in Fig. 4, indicate that the baseline solution leads to an MSE of $2.74e^{-4}$ on the training set and an MSE of $2.86e^{-4}$ on the test set. The GNN architecture rapidly captures correlations in the data and reaches an error of $4.88e^{-6}$ MSE on the training set and $4.36e^{-6}$ MSE on the test set in just 40 training epochs. The RGNN architecture requires 5X more epochs to stabilize on the test set MSE with an error of $2.75e^{-5}$ MSE on the training set and $2.47e^{-5}$ MSE on the test set. However, note that the RGNN model attempts to predict 50 steps into the future—a drastically more difficult task compared to the single time step prediction; thus, a higher error is reasonable and expected for the RGNN model. In the presented MSE metric, we assign the same weight to all nodes, albeit the behavior of nodes far from the laser can be trivial. With this in mind, one can devise alternative metrics to put more focus on nodes with more challenging behavior. As an example of such a metric, the evolution of training and test losses only on 30% of the nodes with the highest temperature is provided in Appendix C - Fig. 10.

We further demonstrate the output of the developed data-driven models for a sample case in the test set (see Fig. 4B). The output of each trained model is compared to the ground truth simulation results for three randomly selected points on the geometry surface for 50 time-steps. This is possible because for all models, although being trained on a fixed number of time steps (1 and 50 respectively), can be recursively evaluated for any arbitrary length of time. The baseline results in a $4.49e^{-4}$ MSE, the GNN model in $3.57e^{-5}$ MSE, and the RGNN model in $5.32e^{-5}$ MSE averaged over all nodes of the simulation. Qualitatively, the results show a good agreement between both GNN and RGNN models and the ground truth; however, the baseline solution falls short of accurately capturing the thermal behavior. Thus, the temporal aspect of the thermal modeling is an essential component of the data-driven modeling. The error values are summarized in Table 2.

To further investigate the stability and capability of the model for long simulations, the models are evaluated on 55 samples (45 for training and 10 for test sets) over 1000 time steps, which is 1000X and 20X the training span of the GNN and RGNN models, respectively (see Fig. 5). Often, such intense extrapolations fail in machine learning; however, we see that both models are capable of reasonable predictions over long periods of time and their errors, although raising over time, remain stable. In Fig. 5A–C, we demonstrate a sample case study for both GNN and RGNN models over 1,000 time steps including their thermal contours as well as root mean squared errors (RMSE) for all material points in each data-driven simulation. Our results indicate that the RGNN model significantly outperforms the GNN model, showing a superior capability to capture long interactions. A similar conclusion can be drawn by observing the RMSE evolution over all training and test samples as shown in Fig. 5D where the GNN model results in a RMSE of $1.5e^{-2}$ for the training set and $1.44e^{-2}$ for the test set, while the RGNN model shows a small error propagation with a RMSE of $9.58e^{-3}$ on the training set and $9.32e^{-3}$ on the test set (see Table 3 for a summary of the results). Both models result in close performance between the training sets and test sets, which shows that they can generalize well across completely unseen geometries for long simulations. A comprehensive view of the results of all test samples over 1000 time steps is provided in Appendix D - Figs. 11 and 12 for the RGNN and GNN models, respectively.

Finally, we discuss the computational costs in the training and deployment process. In our case, the database creation from FEM simulations takes 2 weeks, including 0.5 weeks of simulation design and 1.5 weeks of FEM execution time. The training process takes approximately 3 and 6 weeks on an Nvidia RTX 8000 GPU for the GNN and RGNN, respectively. However, once trained the data-driven model performs far

faster than the FEM counterpart. For example, the RGNN model takes on average 20 s to run the entire simulation in the database, whereas the average execution time for FEM database is 3 hours. This means that a data-driven model can be far more suitable where there is a need for iterative evaluation, such as optimization, robust design, and uncertainty quantification.

4. Conclusions and future works

Recent advancements in high-throughput computing combined with the popularity of data-sharing protocols and cyber-physical systems create a unique opportunity to develop data-driven models for heterogeneous materials and challenging manufacturing processes especially AM processes. Here, we address a key gap in the capability of state-of-the-art data-driven AM models related to their poor generalizability across geometries. We demonstrate that our proposed RGNN architecture effectively captures local intricacies of the process through a graph representation and long-term temporal correlations via a recurrent network structure. This achieves unprecedented generalizability over unseen geometries and maintain it through 1000 time steps, which is over 20X of its training span.

Further improvement can be easily achieved by expanding on both key elements of the data-driven models: database and network. As the model heavily depends on the size and quality of the database, an improvement avenue is to broaden the database to different process parameters, materials, and geometries. Similarly, one can improve the network by deploying larger networks (as we do not observe overfitting in our models) and training over longer time step periods to further reduce the errors. While the presented work trains on simulation data, this framework can be deployed directly on experimental data as high-quality shared repositories for AM processes are growing. As another future research direction, this approach opens the path for new generations of physics-informed data-driven modeling in AM processes with the flexibility to go beyond thermal responses and also predict other challenging aspects of AM processes, such as residual stresses and porosities.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Jian Cao reports financial support was provided by US Department of Defense. Jian Cao reports financial support was provided by National Institute of Standards and Technology. Kornel F. Ehmann reports financial support was provided by National Science Foundation. Mojtaba Mozaffar reports financial support was provided by Department of Defense. Shuheng Liao reports financial support was provided by National Institute of Standards and Technology.

Acknowledgement

This work was supported by the Vannevar Bush Faculty Fellowship N00014-19-1-2642, National Institute of Standards and Technology (NIST) – Center for Hierarchical Material Design (CHiMaD) under grant No. 70NANB14H012, and the National Science Foundation (NSF) – Cyber-Physical Systems (CPS) under grant No. CPS/CMMI-1646592.

Appendix A. – Network block diagrams and training details

See Figs. 6 and 7.

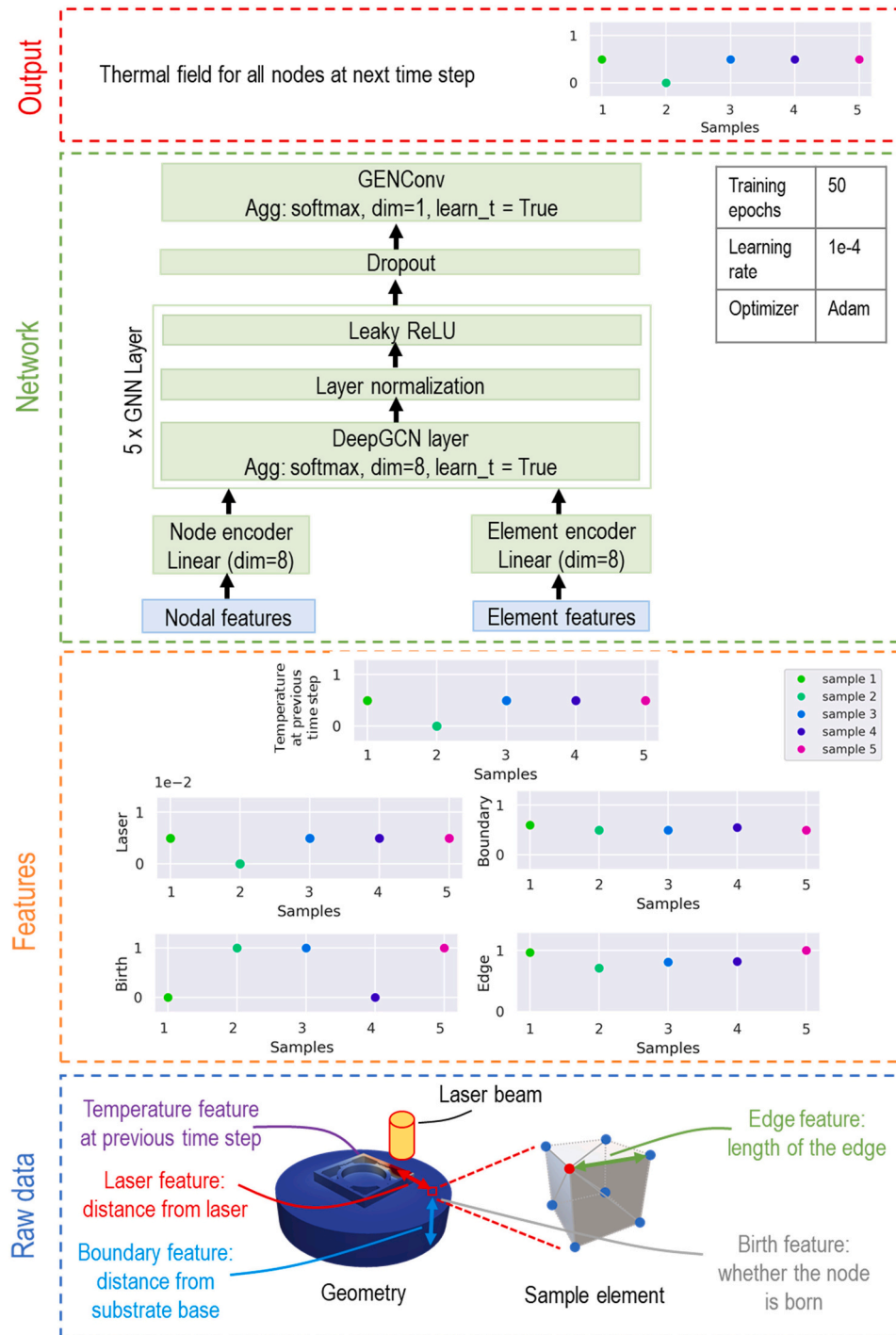


Fig. 6. GNN network architecture, inputs features, outputs and key hyperparameters.

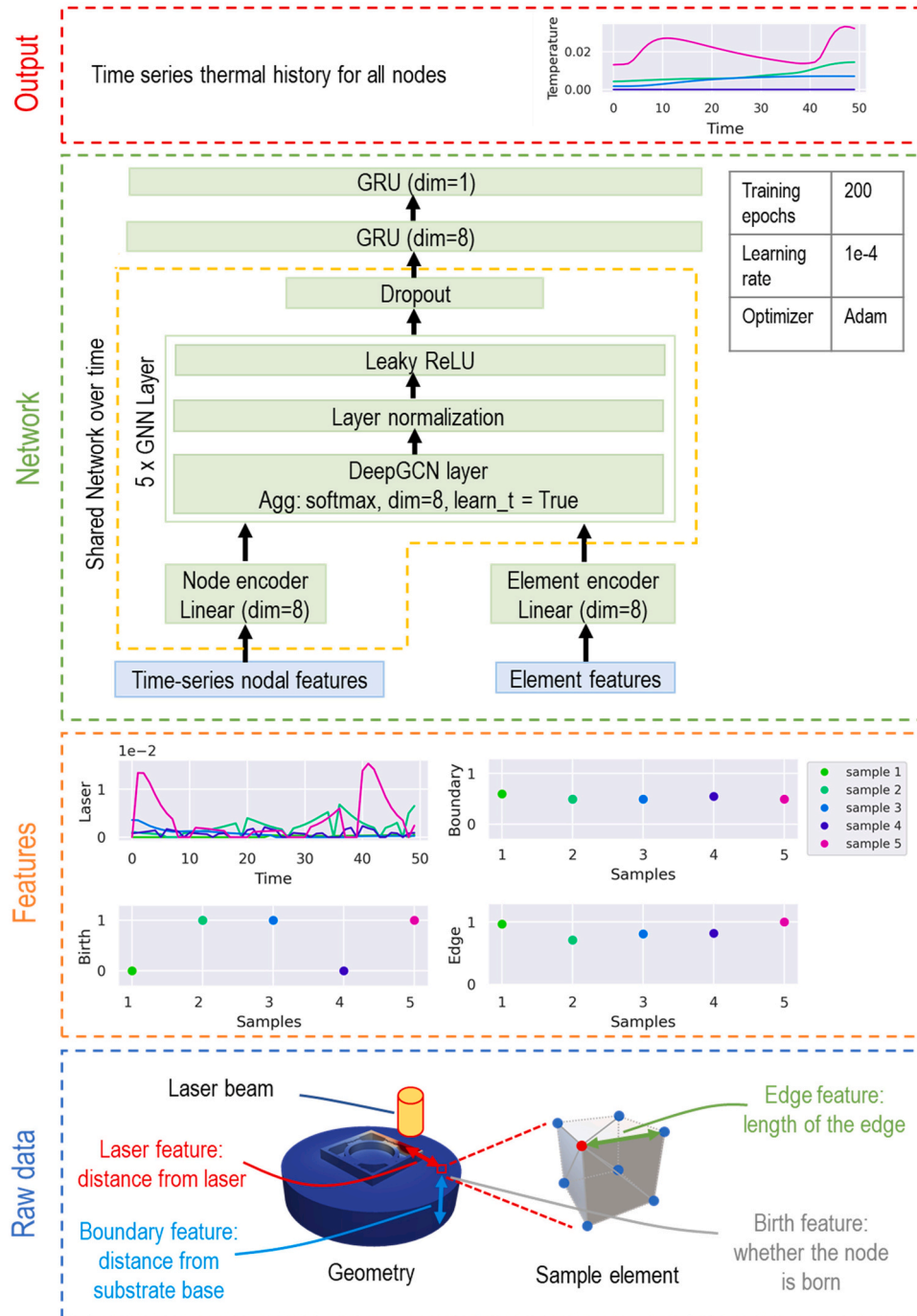


Fig. 7. RGNN network architecture, inputs features, outputs and key hyperparameters.

Appendix B. – Database geometries and characteristics

See Figs. 8 and 9.

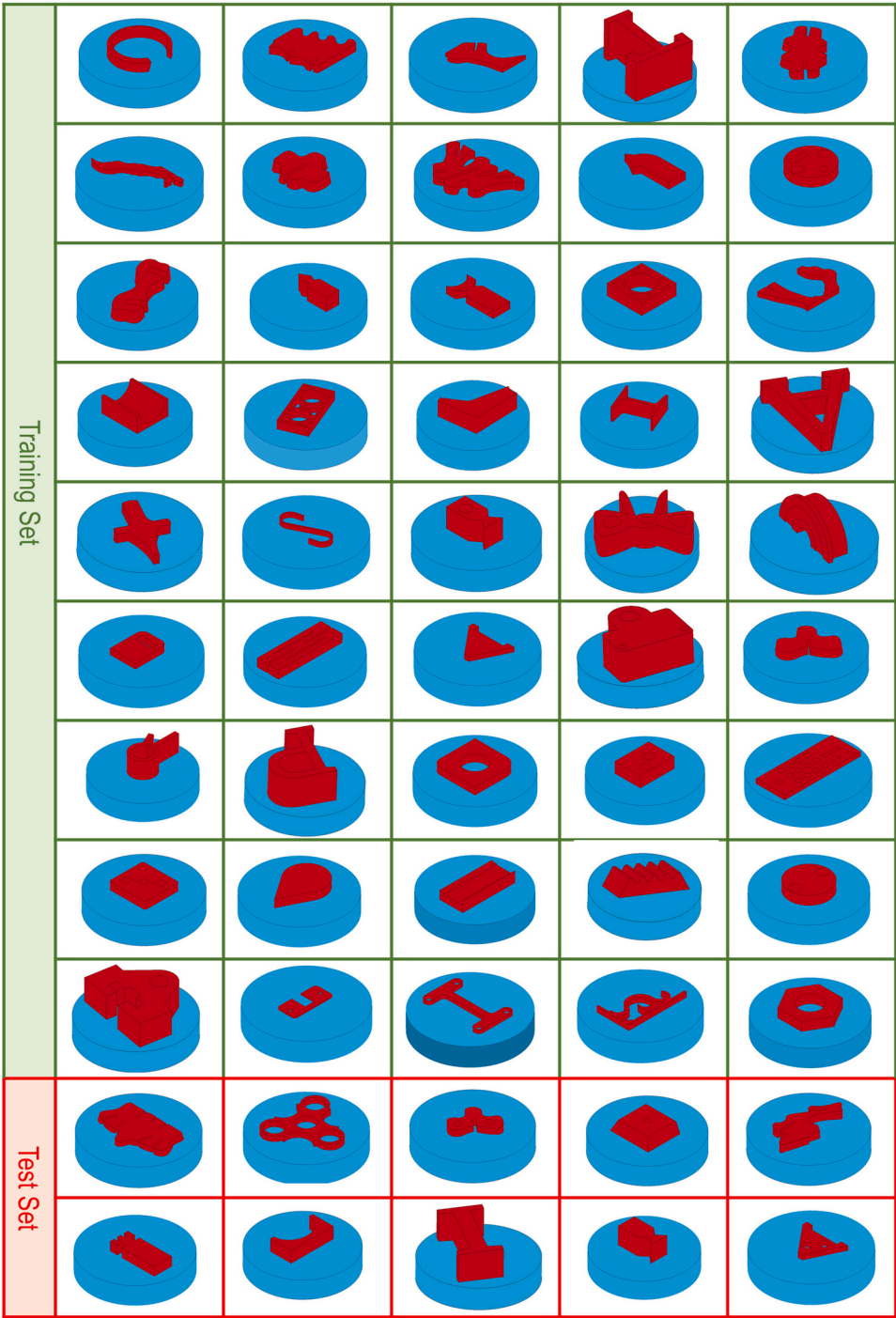


Fig. 8. List of geometries used to produce training and test sets.

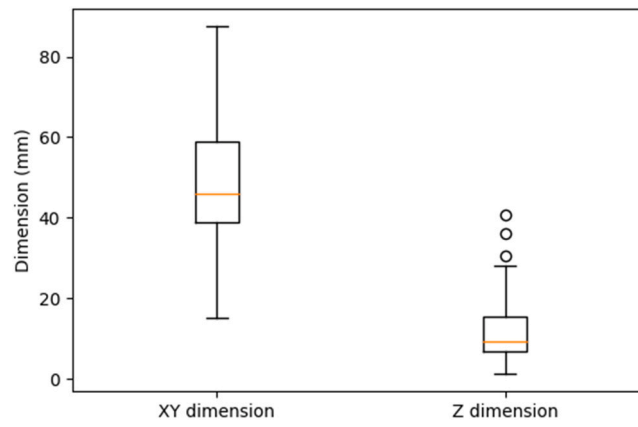


Fig. 9. Database size statistics in XY and Z dimensions.

Appendix C. – Alternative metric

See Fig. 10.

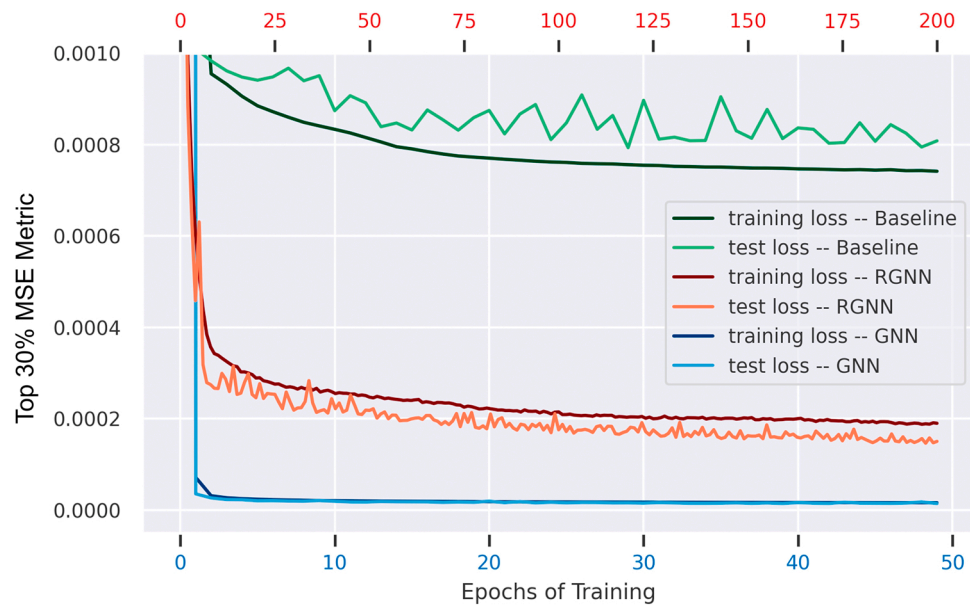


Fig. 10. The evolution of the train and test an alternative metric over training epochs for baseline, GNN, and RGNN models. The metric is computing as the MSE between the predicted and database temperatures at nodes in the top 30% thermal range in any given time step.

Appendix D. – Test-Set results

See Figs. 11 and 12.

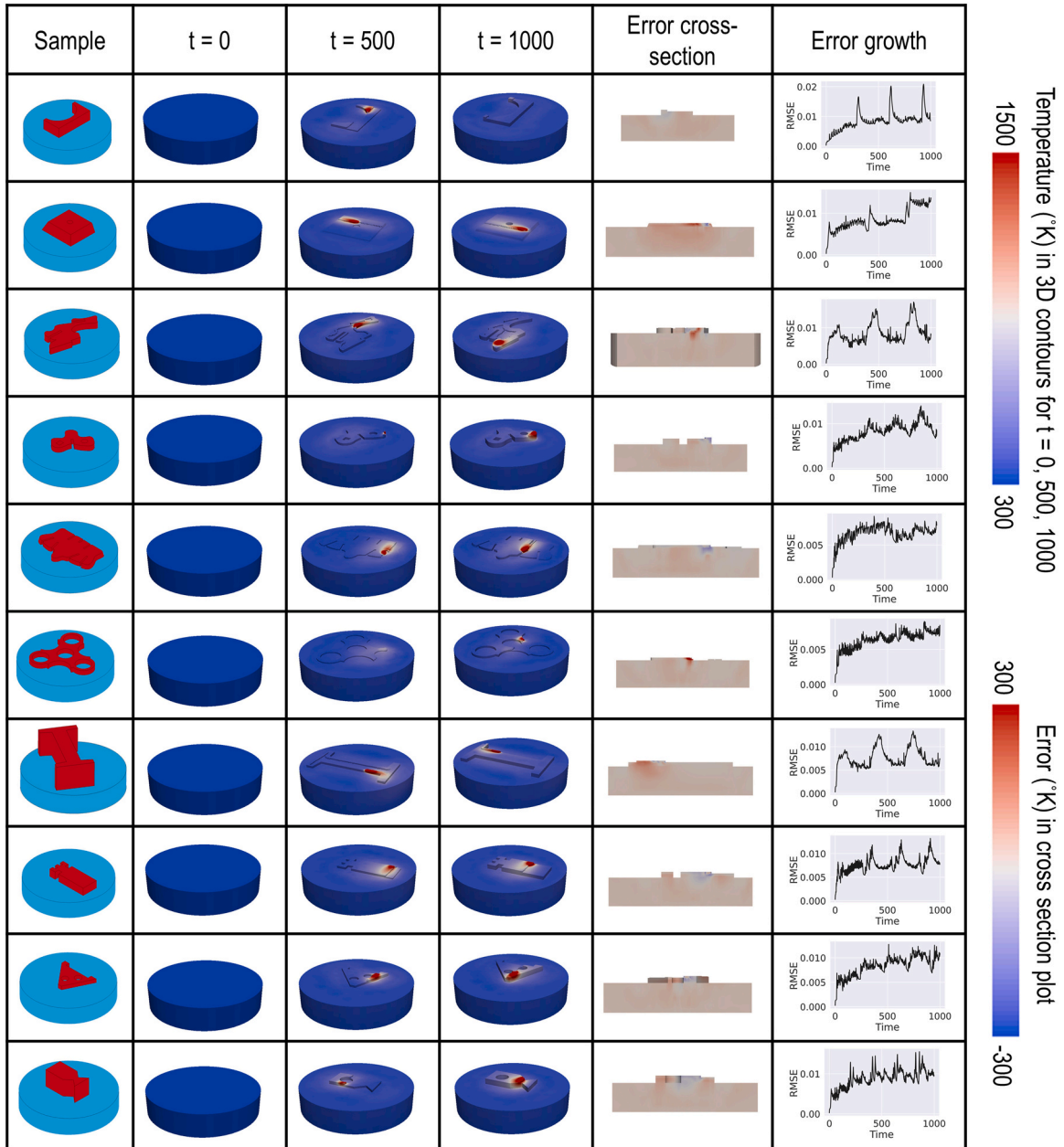


Fig. 11. Results of all test-set geometries with pretrained RGNN network for 1000 time steps including 3D contours at time 0, 500, and 1000 (columns 2–4), cross-section plot of the prediction error at the location of the laser (column 5), and the error growth over time averaged over all nodes (column 6).

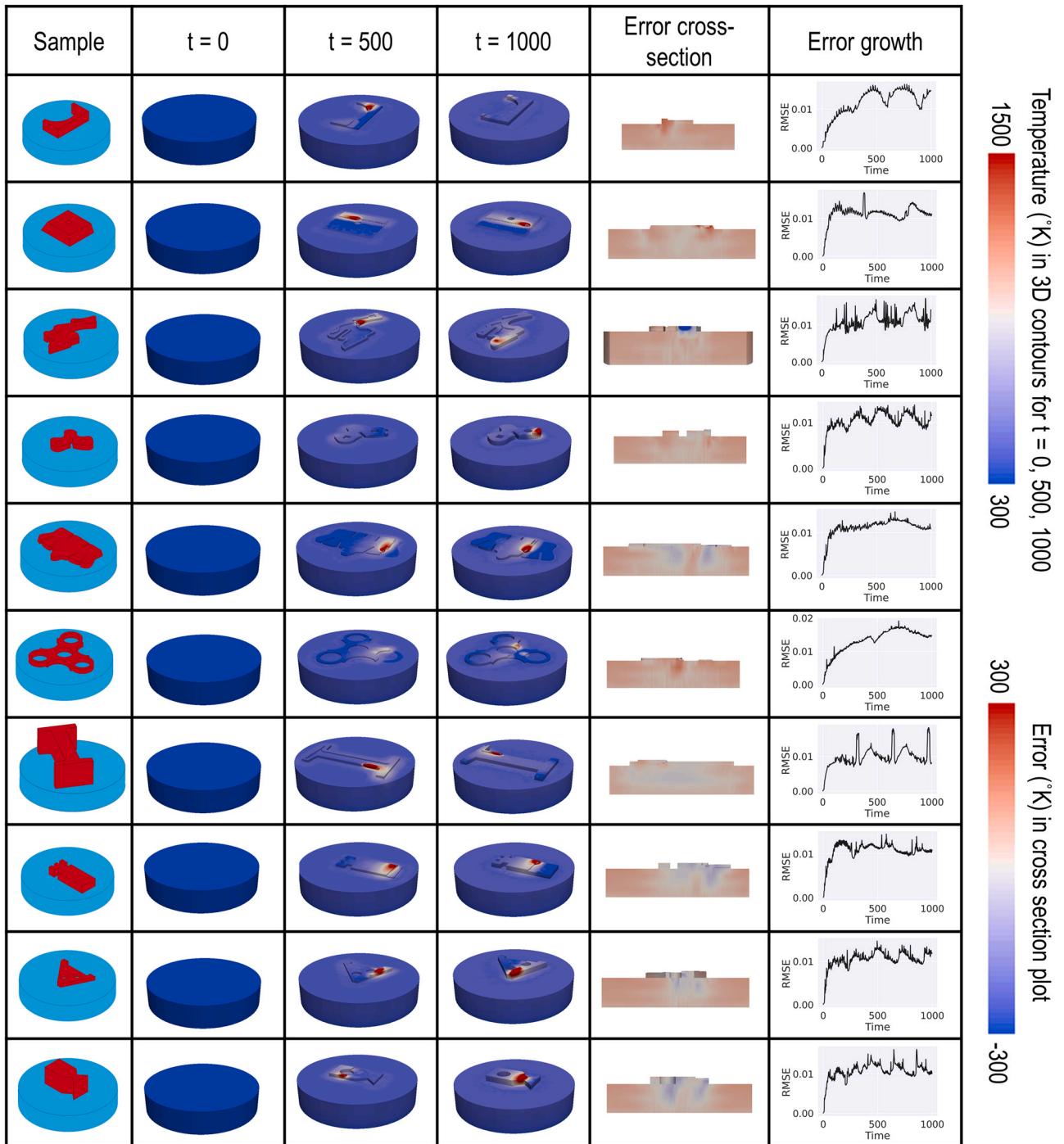


Fig. 12. Results of all test-set geometries with pretrained GNN network for 1000 time steps including 3D contours at time 0, 500, and 1000 (columns 2–4), cross-section plot of the prediction error at the location of the laser (column 5), and the error growth over time averaged over all nodes (column 6).

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